Homework Set 9 - Synthea

Purpose: We are revisiting and old dataset, Synthea, but this time the formatting is different. Before the data was structured with the relational model. Now it is stored in the document model. This assignment is all about working with data in a different model and system. You will be solving the same challenges as before and paying close attention to what is easier or harder (or seems harder because the document model is less familiar). This work is a step on the journey in expanding your mind beyond the flat file world.

Instructions: Put your solutions for all problems in a google colaboratory notebook. Create a level-1 header for each problem and then present the solution in subsequent cells. The header must contain the problem number followed by "confidence x" where x is an integer from 1 to 10 with 10 representing high confidence. For full credit you must show your work. The submission is made via a link on UVACanvas.

Domain Extras: We will be working with the data from the Synthea project. You have seen this before in HW 5. Now we see this data again in its true form, FHIR. There is no new domain information because you have already worked with this data. However you may want to reaquaint yourself.

Data

The data is available from the synthea downloads site `https://synthea.mitre.org/downloads`. Use the file linked behind "1K Sample Synthetic Patient Records, FHIR R4 ".

9.1 Pipeline: Preparation (10ψ)

Load the data into Mongo Atlas (see DS 2022 Lab 9). (If there is more data than fits into your Atlas, that is ok, just put in what you can).

Show code for connecting to the database. Use `pymongo` for this problem and subsequent problems.

9.2 Pipeline: Analysis (10ψ)

Print out the patient name and out-of-pocket cost for medication for each patient with the acute bronchitis condition.

Print the total number of patients and total out-of-pocket costs.

9.3 Pipeline: Analysis (10ψ)

Print out the patient name and total duration of all care plans for each patient with the acute bronchitis condition.

Print the total number of patients and total duration of all care plans.

9.4 Pipeline: Visualization (10ψ)

Make a chart showing the histogram of total amount paid on each claim.

Make a second chart for total amount paid up to $80,000.

9.5 Focus: Press Release (10ψ)

Explore the data dictionary and create your own query (have a little fun, be an investigator). Then create a press release explaining your finding. Make sure it includes:

- Headline
- Statement of why this is important
- Detailed description of the finding
- Visualization explaining the finding

9.6 Focus: Compare (10ψ)

Revisit HW-5 (if you haven't already) and study how you completed the analysis and visualization tasks using the relational model.

Write two paragraphs comparing and contrasting the work you did completing the analysis and visualization tasks with the two different databases.

9.7 Project 2: (30ψ)

For this section all items are based on your specific project.

Problem Definition

(3ψ) Item 1. State the initial general problem and refined specific problem statement

(3ψ) Item 2. One paragraph explaining the motivation for the project

(3ψ) Item 3. One paragraph explaining the rationale for that refinement

(1ψ) Item 4. Headline of Press Release and link to press release (use an anchor link to where it appears in your notebook)

Domain Exposition

(2ψ) Item 1. Terminology - Table summarizing jargon, KPIs, etc.

(2ψ) Item 2. Paragraph explaining the domain the project lives in.

(5ψ) Item 3. Background reading Separate folder containing copies of articles, blog posts, etc. about the project topic (1 point per, up to 5 points, can do more). These should help the reader understand the domain of the project.

(1ψ) Item 4. Table - showing a summary of the readings, one row per ritem, includes title, brief description, and link to file in folder

Press Release

(2ψ) Item 1. Headline - L1 header

(2ψ) Item 2. Hook - L2 header - Short, holds attention, explains the motivation

(2ψ) Item 3. Problem Statement - L2 header - Describe the current state of affairs in more detail and explain your specific problem

(2ψ) Item 4. Solution Description - L2 header - Explain your solution but keep it high level, the target is the end-user not a technical person

(2ψ) Item 5. Chart - L2 Header - A chart that visualizes your data and supports your solution

9.8 Final Prep: (10ψ)

Short answer: use one paragraph to answer the following question.

Q: When sharing a dataset why is it essential to share the metadata as well?